COE 311K Midterm Project

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1 Introduction

Through the use of mathematics, a data set can be interpreted and analyzed in many different ways. This report will go over some of those methods to understand an excerpt dataset from the US Bureau of Economic Analysis.

Quarter	GDP Growth (%)	Quarter	GDP Growth (%)
2010 Q1	1.7	2017 Q1	1.3
2011 Q1	0.1	2018 Q1	2.5
2012 Q1	2.3	2019 Q1	3.1
2013 Q1	2.7	2020 Q1	−5.1
2014 Q1	1.7	2020 Q2	−28.1
2014 Q3	5.0	2020 Q3	33.8
2015 Q2	3.0	2021 Q1	6.3
2016 Q1	1.5	2022 Q1	−1.6
2016 Q3	3.5	2023 Q2	2.4
2016 Q4	1.8	2023 Q4	3.3

Figure 1: Excerpt Data set

The methods that will be used on this data will be cubic spline interpolation and polynomial fitting using least squares method. Throughout this report, the effects of these methods will be explained in the context of this data, with an additional discussion of residuals and error associated with each method.

2 Part A: Cubic Spline Interpolation

The process of constructing a natural cubic spline through a data set is to define a cubic polynomial between each set of consecutive data points. Putting together this piecewise function provides us with the final spline.

2.1 Question 1

2.1.1 Unknowns

We start with n unknowns, where n is the number of data points. It is a fact that the end points of the polynomial will have a second derivative of zero,

which we use to our advantage. This leaves us with n-2 unknowns.

2.1.2 System of Equations

With the endpoints known, we have only the interior points to solve for. We can refer to each of these points as follows:

$$i = 1, 2, ..., n - 2 \tag{1}$$

This allows us to form the following equation, while also defining the width of each interval.

$$h_{i-1}M_{i-1} + 2(h_{i-1} + h_i)M_i + h_iM_{i+1} = 6\left(\frac{y_{i+1} - y_i}{h_i} - \frac{y_i - y_{i-1}}{h_{i-1}}\right) \tag{2}$$

$$h_i = x_{i+1} - x_i \tag{3}$$

This produces a tridiagonal linear system, where the lower, main, and upper diagonals make up the above equation.

2.2 Continuity Conditions

Since we are working with multiple smaller segments, using continuity conditions ensures that the final spline is cohesive. At each interior knot, it must be continuous along the function, its first derivative, and its second derivative.

$$S_i(x_i) = S_{i-1}(x_i) \tag{4}$$

$$S_i'(x_i) = S_{i-1}'(x_i) \tag{5}$$

$$S_i''(x_i) = S_{i-1}''(x_i) \tag{6}$$

2.3 Boundary Conditions

For a natural cubic spline, we assume that there is no curvature at the endpoints, and therefore, the second derivative at these points is zero.

2.3.1 Graph and Error

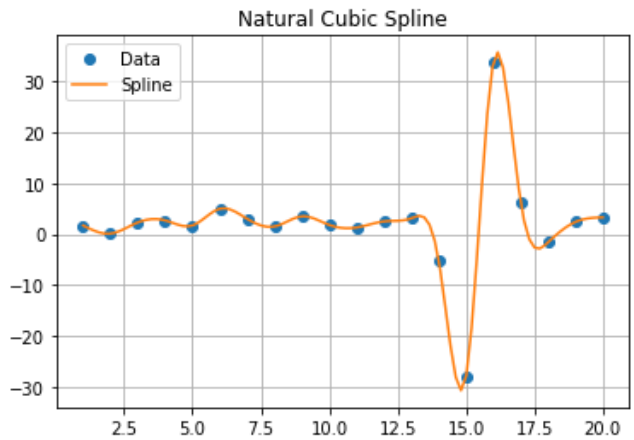


Figure 2: Natural Cubic Spline

Since the data points have some distance between them, this graph is made up of 100 x values in between the original range. These values are then evaluated with the natural cubic spline process described above. This allows for smooth graphing.

This process of spline creation works to connect all the data points. As such, the error values are as follows:

Mean Squared Error: 0
Residual Squared Value: 1

This makes sense since the function inherently incorporates each data point. This can be visualized since all the data points fall on the line in Figure 2.

2.4 Question 2

It was previously mentioned that the provided data is an excerpt from a larger data set. This poses the question, using the cubic spline interpolation method and the provided data, can we extrapolate the data for the remaining quarters within this range?

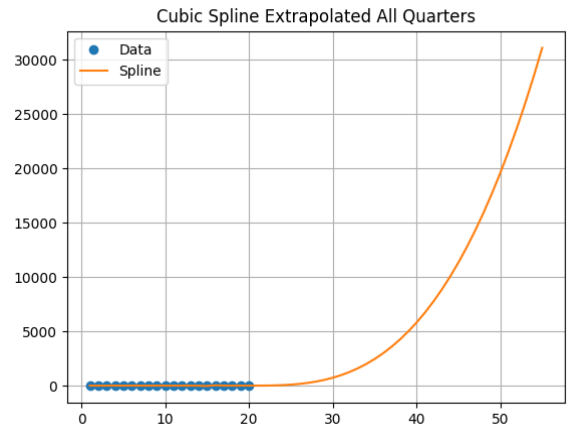


Figure 3: Cubic Spline Extrapolation

2.4.1 Graph and Analysis

It can be seen that beyond the given data, the spline explodes. It is unable to accurately extrapolate for all quarters. With given x and y points, the algorithm does a good job of capturing the data. However, past the given GDP values (y values), the loop within the code finishes without a match. This means the previous slope is used continuously and the function keeps growing.

As the name suggests, the natural cubic spline algorithm is a beautiful interpolation method but is not equipped to understand patterns and predict values.

2.5 Question 3

The Runge Phenomenon occurs when interpolation is forced through every present point, including extreme outliers (such as the 2020 Q2 value). This can cause extreme curvature changes on either side of the outlier. Additionally, it causes unexpected oscillations and can model the system incorrectly.

2.5.1 Trade-Offs

There exists a trade-off between exact interpolation and stability. Interpolating through every data point ensures an accurate understanding of the data. In situations like this, the outlier is significant because Covid greatly effected the GDP. However, sometimes data is noisy and outliers are insignificant. In such cases, it is beneficial to use approximation to produce smoother graphs, meaning more stable models.

This scenario, the least squares approach is preferable. Smoothing the spline would reduce the small differences between the quarters, and reduce the impact that Covid had on the GDP. It was an important crisis that had significant economic effect and its magnitude shouldn't be diminished. This contrasts the least squares method, which captures the trend without forcing interpolation through all the points.

3 Part B: Polynomial and Least Squares Comparison

3.1 Question 1

Another method to analyze the data is to create a polynomial of nth degree using the least squares method. This is done by creating a system of equations for every point.

$$(x_0, y_0) : a_n x_0^n + ... + a_1 x_0 + a_0 = y_o \tag{7}$$

These equations then come together to form the vandermonde matrix (V) in the following form.

$$Va = y \tag{8}$$

The coefficients can be solved by manipulating the equation into the following form.

$$a = V^{-1} y \tag{9}$$

3.1.1 Graphs

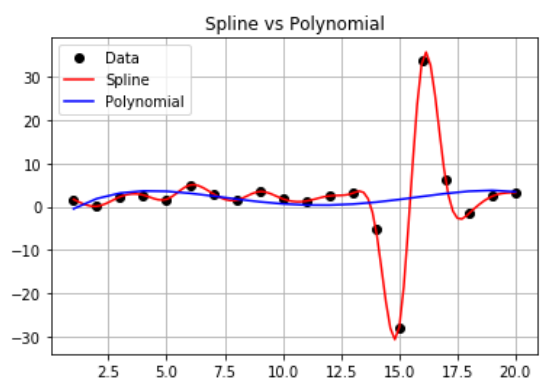


Figure 4: Polynomial vs Cubic Spline

The above graph demonstrates the difference between using cubic spline interpolation (red) and a 4th degree polynomial approximation (blue). The polynomial clearly smoothens the data and disregards the outlier caused by Covid.
This is the residuals plot for both of the lines from Figure 4. It can be seen that

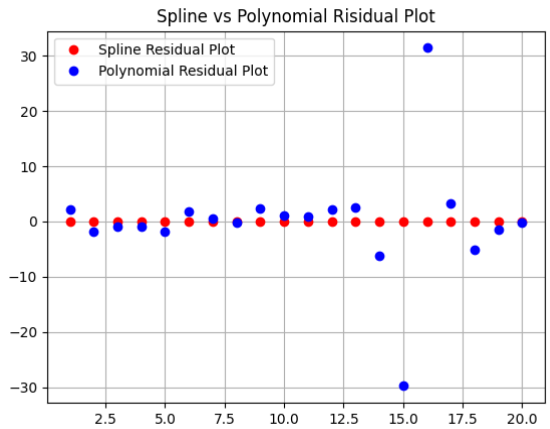


Figure 5: Polynomial vs Cubic Residuals

the Cubic Spline residuals are all zero since interpolation forces the spline to go through every point. This contrasts the 4th degree polynomial that shows some variance around the data, including large residuals for the covid affected data points.

3.1.2 Error

The error for the 4th degree polynomial approximation is as follows.

Mean Squared Error: 99.3
Residual Squared Value: 0.02

These values are far from the previous calculated error for the cubic spline interpolation. The large MSE means that the model's predictions for GDP values are significantly off compared to the actual data set. The small R^2 value means that the model explains very little of the variance in the data. However, we can visually see that most of the graph is smooth. These error values are being significantly influenced by the impact that Covid had on the GDP. The model is not able to account for that, but captures the overall trend fairly well.

3.1.3 Analysis

Since the impact that Covid had on the GDP is not an outlier that is expected repeatedly, the polynomial approximation is good at capturing the overall trend. However, if the desire is to reproduce specific data points, the cubic spline does a better job since it contains all of the original data.

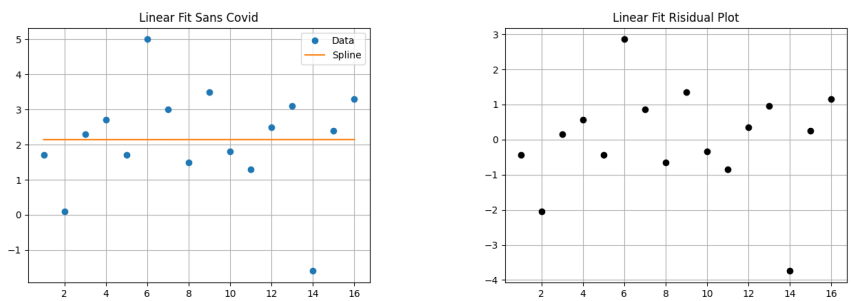
The approximation allows for decent prediction for future GDP values, but loses details that are important to understanding the economic state during certain

time periods (such as Covid). The cubic spline interpolation is beneficial for obtaining an accurate understanding of trends within data that already exists, and can capture significant economic events (such as Covid), but cannot predict past the scope of the data it is created around.

3.2 Question 2

As seen in the previous question, the data from the quarters relating to Covid increase our error and make it difficult to find a trend. As such, this is a linear test for the remaining data set that removes all the quarters affected by Covid.

3.2.1 Graph



(a) Linear Fit

(b) Residual Plot

Figure 6: Linear Approximation for No Covid Data

3.2.2 Analysis

The graph in Figure 6 shows a horizontal line with a slope of 0. A linear trend does not make sense for this data. The GDP is constantly changing and goes through long term visible oscillation. A linear trend assumes that the changes in GDP are constant and that is not true. The system is always changing in response to world events.

It is important to note that the error with this model demonstrates that this is stronger and more accurate.

Mean Squared Error: 2.08
Residual Squared Value: 0.0

However, using real world intuition, a linear model is not a strong model for GDP analysis or prediction.

4 Part C: Method Justification

4.1 Question 1

4.1.1 Smoothness / Sensitivity to Outliers

The polynomial approximation is better at showing smooth trends and is less sensitive to the effects that outliers have on it. An economy's recovery to an off quarter (essentially an outlier in the data) may not be as oscillatory as a cubic spline might make it seem.

4.1.2 Accuracy

While cubic spline interpolation ensures accuracy throughout a given data set, it is not helpful for predictions. The model explodes rather than being able to understand and predict a pattern in the data. For a policymaker who is estimating GDP growth, they need a model that can accurately predict based on existing patterns, which the polynomial approximation can do.

4.1.3 Cost

While the polynomial approximation is beneficial for predictions based on trends, it is more computationally expensive. Complexity for cubic spline interpolation is O(n) vs for polynomial fitting is $O(n^3)$

4.1.4 Recommendation

Despite the polynomial approximation being more computationally expensive, it is the more desirable method for the policymaker.

Qualitative: The smoothness of the graph and the reduced sensitivity to outliers produces a more stable model.

Quantitative: The ability to predict future values is beneficial to the policymaker.

5 Conclusion

Cubic spline interpolation and polynomial approximation are both ways to interpret data. Cubic spline interpolation works well to capture the true essence of a data set, ensuring that all points (including meaningful outliers) are accounted. However, this model is not equipped with the ability to predict past its given scope. This makes cubic spline interpolation the right model to gain an accurate understanding of a data set through interpolation. On the other hand, polynomial approximation is a powerful prediction model. It creates a smoother graph that looks at the overall trend and can be used to extrapolate beyond the scope of the data. However, this means that outliers have less impact, and it is possible to lose an accurate understanding of certain events (like Covid).

To summarize, polynomial approximation takes much longer to compute when given a larger data set to analyze, but is better at identifying trends and predictions. Cubic spline interpolation captures each point and is computationally less expensive, but is not a good prediction model.